

Homework 3

Submission 1, Spring 2026

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```
#| include: false

options(repos = c(CRAN = "https://cloud.r-project.org"))

if (!require("pacman")) install.packages("pacman") pacman::p_load(tidyverse, ggplot2, dplyr,
lubridate, stringr, readxl, data.table, gdata, scales, data.table)

ma.2010 <- read_csv("../data/output/data-2010.csv") ma.2011 <- read_csv("../data/out-
put/data-2011.csv") ma.2012 <- read_csv("../data/output/data-2012.csv") ma.2013 <-
read_csv("../data/output/data-2013.csv") ma.2014 <- read_csv("../data/output/data-
2014.csv") ma.2015 <- read_csv("../data/output/data-2015.csv")

fix_type <- function(df) { dforgparent <- -as.character(dforg_parent) df } ma.2011 <-
fix_type(ma.2011)

data.full <- bind_rows(ma.2010, ma.2011, ma.2012, ma.2013, ma.2014, ma.2015) %>% mu-
tate(market_share = avg_enrollment / avg_enrolled)

summary_table <- data.full %>% group_by(year) %>% summarise( 'Average Star Rating' =
round(mean(Star_Rating, na.rm = TRUE), 2), 'Average Enrollments' = round(mean(avg_en-
rollment, na.rm = TRUE), 2), 'Average Market Share' = round(mean(market_share, na.rm =
TRUE), 4), 'Total Number Of Plans' = n() )

print(summary_table)

without_rating_table <- data.full %>% filter(is.na(Star_Rating)) %>% group_by(year)
%>% summarise( 'Average Enrollments' = round(mean(avg_enrollment, na.rm = TRUE), 2),
'Average Market Share' = round(mean(market_share, na.rm = TRUE), 4), 'Total Number Of
Plans' = n() )

print(without_rating_table)

data.full %>% filter(year %in% c(2010, 2012, 2015), !is.na(Star_Rating)) %>% mutate(year =
as.factor(year)) %>% ggplot(aes(x = as.factor(Star_Rating))) + geom_bar() + facet_wrap(~
year) + labs( title = "Distribution of Star Ratings over 2010, 2012, and 2015", x = "Star
Rating", y = "Number Of Plans" ) + theme_minimal()
```

```

data.full <- data.full %>% mutate( star_3 = as.integer(Star_Rating == 3), star_35 =
as.integer(Star_Rating == 3.5), star_4 = as.integer(Star_Rating == 4), star_45 = as.integer(Star_Rating >= 4.5) )

estimate <- list()

for(yr in 2010:2015){ ols <- lm(market_share ~ star_3 + star_35 + star_4 + star_45, data =
data.full %>% filter(year == yr, !is.na(Star_Rating), !is.na(market_share))) estimate[[as.character(yr)]] <- coef(ols) }

ols_table <- as.data.frame(estimate) colnames(ols_table) <- c("2010", "2011", "2012", "2013",
"2014", "2015") rownames(ols_table) <- c("Excluded (<= 2.5 stars)", "3 Stars", "3.5 Stars",
"4 Stars", ">= 4.5 Stars")

print(round(ols_table, 4))

quality_vars <- c( "breastcancer_screen", "rectalcancer_screen", "cv_diab_cholscreen", "glaucoma_test", "monitoring", "flu_vaccine", "pn_vaccine", "physical_health", "mental_health", "osteo_test", "physical_monitor", "primaryaccess", "osteo_manage", "diab_healthy", "bloodpressure", "ra_manage", "copd_test", "bladder", "falling", "nodelays", "doctor_communicate", "carequickly", "customer_service", "overallrating_care", "overallrating_plan", "complaints_plan", "appeals_timely", "appeals_review", "leave_plan", "audit_problems", "hold_times", "info_accuracy", "ttyp_available" )

library(dplyr)

ma.2010 <- ma.2010 %>% rowwise() %>% mutate( raw_rating = mean(c_across(all_of(quality_vars))), na.rm = TRUE) ) %>% ungroup()

round_up_counts <- ma.2010 %>% filter(!is.na(raw_rating)) %>% filter(!is.na(partc_score))
%>% mutate( round_ups = case_when( raw_rating >= 2.75 & partc_score == 3.0 ~ "Rounded up to 3", raw_rating >= 3.25 & partc_score == 3.5 ~ "Rounded up to 3.5", raw_rating >= 3.75 & partc_score == 4.0 ~ "Rounded up to 4", raw_rating >= 4.25 & partc_score == 4.5 ~ "Rounded up to 4.5", raw_rating >= 4.75 & partc_score == 5.0 ~ "Rounded up to 5", TRUE ~ NA_character_ ) ) %>% filter(!is.na(round_ups)) %>% count(round_ups) %>% rename('Star Rating' = round_ups, 'Corresponding Number Of Plans' = n) print(round_up_counts)

ma_25star_candidates <- ma.2010 %>% filter( !is.na(raw_rating), !is.na(partc_score), Star_Rating %in% c(2.5, 3) )

n_candidates_total <- nrow(ma_25star_candidates) n_candidates_by_score <- ma_25star_candidates %>% count(partc_score)

ma_25star <- ma_25star_candidates %>% filter( raw_rating >= 2.5, raw_rating <= 3, (raw_rating >= 2.75 & Star_Rating == 3) | (raw_rating < 2.75 & Star_Rating == 2.5) )

```

```

ma.rd1 <- ma_25star %>% mutate(market_share = avg_enrollment / avg_enrolled,
score = raw_rating - 2.75, treat = (score>=0), window = (score>=-.125 & score<=.125),
score_treat=score*treat)

library(rdrobust)

star25.1 <- lm(market_share ~ score + treat + score_treat, data= (ma.rd1 %>% filter(win-
dow==TRUE))) est1 <- as.numeric(star25.1$coef[3])

ma_25star_candidates <- ma.2010 %>% filter( !is.na(raw_rating), !is.na(partc_score),
Star_Rating %in% c(3, 3.5) )

n_candidates_total <- nrow(ma_25star_candidates) n_candidates_by_score <-
ma_25star_candidates %>% count(partc_score)

ma_25star <- ma_25star_candidates %>% filter( raw_rating >= 3, raw_rating <= 3.5,
(raw_rating >= 3.25 & Star_Rating == 3.5) | (raw_rating < 3.25 & Star_Rating == 3) )

ma.rd2 <- ma_25star %>% mutate(market_share = avg_enrollment / avg_enrolled,
score = raw_rating - 3.25, treat = (score>=0), window = (score>=-.125 & score<=.125),
score_treat=score*treat)

library(rdrobust)

star25.2 <- lm(market_share ~ score + treat + score_treat, data= (ma.rd2 %>% filter(win-
dow==TRUE))) est2 <- as.numeric(star25.2$coef[3])

```

our two estimates

```

est1 <- as.numeric(star25.1$coef[3]) est2 <- -as.numeric(star25.2$coef[3])

comparison_table <- data.frame(row.names = c("RD Estimate"), 3 vs 3.5 = est1, 2.5 vs 3
= est2 )

print(comparison_table)

ma_25star_candidates <- ma.2010 %>% filter( !is.na(raw_rating), !is.na(partc_score),
Star_Rating %in% c(2.5, 3) )

n_candidates_total <- nrow(ma_25star_candidates) n_candidates_by_score <-
ma_25star_candidates %>% count(partc_score)

ma_25star <- ma_25star_candidates %>% filter( raw_rating >= 2.5, raw_rating <= 3,
(raw_rating >= 2.75 & Star_Rating == 3) | (raw_rating < 2.75 & Star_Rating == 2.5) )

ma.rd1 <- ma_25star %>% mutate(market_share = avg_enrollment / avg_enrolled, score =
raw_rating - 2.75, treat = (score>=0), window1 = (score>=-.1 & score<=.1), window2 =

```

```
(score>=-.12 & score<=.12), window3 = (score>=-.13 & score<=.13), window4 = (score>=-.14 & score<=.14), window5 = (score>=-.15 & score<=.15), score_treat=score*treat)
```

```
library(rdrobust)
```

```
star25.1 <- lm(market_share ~ score + treat + score_treat, data= (ma.rd1 %>% filter(window1==TRUE))) star25.2 <- lm(market_share ~ score + treat + score_treat, data= (ma.rd1 %>% filter(window2==TRUE))) star25.3 <- lm(market_share ~ score + treat + score_treat, data= (ma.rd1 %>% filter(window3==TRUE))) star25.4 <- lm(market_share ~ score + treat + score_treat, data= (ma.rd1 %>% filter(window4==TRUE))) star25.5 <- lm(market_share ~ score + treat + score_treat, data= (ma.rd1 %>% filter(window5==TRUE))) est1 <- as.numeric(star25.1$coef[3]) est2 <- -as.numeric(star25.2$coef[3]) est3 <- as.numeric(star25.3$coef[3]) est4 <- -as.numeric(star25.4$coef[3]) est5 <- as.numeric(star25.5$coef[3])
```

```
library(ggplot2)
```

Create a dataframe of estimates and bandwidths

```
rd_results <- data.frame( bandwidth = c(0.10, 0.12, 0.13, 0.14, 0.15), estimate = c(est1, est2, est3, est4, est5) )
```

Add confidence intervals

```
se_list <- c( summary(star25.1)$coef[3, 2], summary(star25.2)$coef[3, 2], summary(star25.3)$coef[3, 2], summary(star25.4)$coef[3, 2], summary(star25.5)$coef[3, 2] )
```

```
rd_results <- rd_results %>% mutate( se = se_list, ci_lo = estimate - 1.96 * se, ci_hi = estimate + 1.96 * se )
```

Plot

```
ggplot(rd_results, aes(x = bandwidth, y = estimate)) + geom_point(size = 3, color = "steelblue") + geom_line(color = "steelblue") + geom_errorbar(aes(ymin = ci_lo, ymax = ci_hi), width = 0.003, color = "steelblue") + geom_hline(yintercept = 0, linetype = "dashed", color = "red") + scale_x_continuous(breaks = rd_results$bandwidth) + labs( title = "RD Estimates at 2.5-Star Threshold", subtitle = "Effect of 3-Star Rating on Market Share", x = "Bandwidth", y = "Estimated Treatment Effect" ) + theme_minimal()
```

```
ma_25star_candidates <- ma.2010 %>% filter( !is.na(raw_rating), !is.na(partc_score), Star_Rating %in% c(3, 3.5) )
```

```

n_candidates_total <- nrow(ma_25star_candidates)  n_candidates_by_score <-
ma_25star_candidates %>% count(partc_score)

ma_25star <- ma_25star_candidates %>% filter( raw_rating >= 3, raw_rating <= 3.5,
(raw_rating >= 3.25 & Star_Rating == 3.5) | (raw_rating < 3.25 & Star_Rating == 3) )

ma.rd2 <- ma_25star %>% mutate(market_share = avg_enrollment / avg_enrolled, score =
raw_rating - 3.25, treat = (score>=0), window1 = (score>=-.1 & score<=.1), window2 =
(score>=-.12 & score<=.12), window3 = (score>=-.13 & score<=.13), window4 = (score>=-.14
& score<=.14), window5 = (score>=-.15 & score<=.15), score_treat=score*treat)

library(rdrobust)

star25.1 <- lm(market_share ~ score + treat + score_treat, data= (ma.rd2 %>% filter(win-
dow1==TRUE))) star25.2 <- lm(market_share ~ score + treat + score_treat, data=
(ma.rd2 %>% filter(window2==TRUE))) star25.3 <- lm(market_share ~ score + treat +
score_treat, data= (ma.rd2 %>% filter(window3==TRUE))) star25.4 <- lm(market_share
~ score + treat + score_treat, data= (ma.rd2 %>% filter(window4==TRUE))) star25.5
<- lm(market_share ~ score + treat + score_treat, data= (ma.rd2 %>% filter(win-
dow5==TRUE))) est1 <- as.numeric(star25.1$coef[3])est2 <- -as.numeric(star25.2$coef[3])
est3 <- as.numeric(star25.3$coef[3])est4 <- -as.numeric(star25.4$coef[3]) est5 <- as.nu-
meric(star25.5$coef[3])

library(ggplot2)

```

Create a dataframe of estimates and bandwidths

```

rd_results <- data.frame( bandwidth = c(0.10, 0.12, 0.13, 0.14, 0.15), estimate = c(est1, est2,
est3, est4, est5) )

```

Add confidence intervals

```

se_list <- c( summary(star25.1)$coef[3, 2], summary(star25.2)$coef[3, 2], summary(star25.3)$coef[3, 2], summary(star25.4)$coef[3, 2], summary(star25.5)$coef[3, 2] )

rd_results <- rd_results %>% mutate( se = se_list, ci_lo = estimate - 1.96 * se, ci_hi =
estimate + 1.96 * se )

```

Plot

```
ggplot(rd_results, aes(x = bandwidth, y = estimate)) + geom_point(size = 3, color =
"steelblue") + geom_line(color = "steelblue") + geom_errorbar(aes(ymin = ci_lo, ymax =
ci_hi), width = 0.003, color = "steelblue") + geom_hline(yintercept = 0, linetype = "dashed",
color = "red") + scale_x_continuous(breaks = rd_results$bandwidth) + labs( title = "RD
Estimates at 3-Star Threshold", subtitle = "Effect of 3.5-Star Rating on Market Share", x =
"Bandwidth", y = "Estimated Treatment Effect" ) + theme_minimal()
```

We note that the findings are indeed sensitive to the choice of bandwidth. For instance, the treatment effect when comparing 3-star versus 3.5-star plans appears more pronounced at a threshold of 0.1 compared to 0.15. Similarly, we observe a difference in the calculated effect when a bandwidth of 0.12 is chosen instead of 0.14 while comparing 2.5-star plans to 3-star plans.

```
ma_25star_candidates <- ma.2010 %>% filter( !is.na(raw_rating), !is.na(partc_score),
Star_Rating %in% c(2.5, 3) )
```

```
n_candidates_total <- nrow(ma_25star_candidates)  n_candidates_by_score <-
ma_25star_candidates %>% count(partc_score)
```

```
ma_25star <- ma_25star_candidates %>% filter( raw_rating >= 2.5, raw_rating <= 3,
(raw_rating >= 2.75 & Star_Rating == 3) | (raw_rating < 2.75 & Star_Rating == 2.5) )
```

```
ma.rd1 <- ma_25star %>% mutate(market_share = avg_enrollment / avg_enrolled,
score = raw_rating - 2.75, treat = (score>=0), window = (score>=-.125 & score<=.125),
score_treat=score*treat) Q8. # Assuming 'window1' is the desired condition for filtering
p1 <- ggplot(ma.rd1 %>% filter(window1 == TRUE), aes(x = score, fill = treat)) +
geom_density(alpha = 0.4) + geom_vline(xintercept = 0, linetype = "dashed", color =
"black", linewidth = 0.8) + scale_fill_manual(values = c("FALSE" = "coral", "TRUE" =
"steelblue"), labels = c("2.5 Stars", "3 Stars")) + labs(title = "Density of 2.5 vs 3 Stars",
x = "Running Variable (Score)", y = "Density", fill = "Rating") + theme_minimal() +
theme(plot.title = element_text(size = 10), # Smaller title axis.title = element_text(size = 8),
# Smaller axis titles legend.title = element_text(size = 8), # Smaller legend title legend.text
= element_text(size = 7)) # Smaller legend text
```

Display the plot

```
print(p1)
```

Assuming ‘window1’ is the desired condition for filtering

```
p2 <- ggplot(ma.rd2 %>% filter(window1 == TRUE), aes(x = score, fill = treat)) + geom_density(alpha = 0.4) + geom_vline(xintercept = 0, linetype = "dashed", color = "black", linewidth = 0.8) + scale_fill_manual(values = c("FALSE" = "coral", "TRUE" = "steelblue"), labels = c("3 Stars", "3.5 Stars")) + labs(title = "Density of 3 vs 3.5 Stars", x = "Running Variable (Score)", y = "Density", fill = "Rating") + theme_minimal() + theme(plot.title = element_text(size = 10), # Smaller title axis.title = element_text(size = 8), # Smaller axis titles legend.title = element_text(size = 8), # Smaller legend title legend.text = element_text(size = 7)) # Smaller legend text
```

Display the plot

```
print(p2)
```

We observe a sharp increase in market share in relation to values on either side of the threshold, especially noticeable when comparing 2.5-star to 3-star plans (as seen in the first plot). Specifically, there is a decline from -0.10 to -0.05 in the upper plot, along with a slight rise from 0.05 to 0.10 in the lower plot.

```
Q9. summary_df <- data.frame( Threshold = c("2.5 vs 3 Stars", "3 vs 3.5 Stars"),  
# Part D counts PartD_Yes = c( sum(ma.rd1 %>% filter(window == TRUE) %>% pull(partd) == "Yes", na.rm = TRUE), sum(ma.rd2 %>% filter(window == TRUE) %>% pull(partd) == "Yes", na.rm = TRUE) ), PartD_No = c( sum(ma.rd1 %>% filter(window == TRUE) %>% pull(partd) == "No", na.rm = TRUE), sum(ma.rd2 %>% filter(window == TRUE) %>% pull(partd) == "No", na.rm = TRUE) ),  
  
# Below/Above threshold Below_Threshold = c( sum(ma.rd1 %>% filter(window == TRUE) %>% pull(score) < 0, na.rm = TRUE), sum(ma.rd2 %>% filter(window == TRUE) %>% pull(score) < 0, na.rm = TRUE) ), Above_Threshold = c( sum(ma.rd1 %>% filter(window == TRUE) %>% pull(score) >= 0, na.rm = TRUE), sum(ma.rd2 %>% filter(window == TRUE) %>% pull(score) >= 0, na.rm = TRUE) ) )
```

print(summary_df) Q9 With a bandwidth of 0.125, we observe that there are significantly more Plan D contracts below the threshold than above, across both categories. Additionally, the number of HMO-approved plans (calculated using the adjusted raw rating, as HMO is defined through Part C) shows considerable variation above and below the threshold. This indicates that there are distinct characteristics among the plans.

Q10 The impact of an increased star rating is directly linked to a higher number of enrollments, especially evident in the upward and downward slopes around the threshold in the Problem 8 plot comparing 2.5-star plans with 3-star plans. However, these findings are highly sensitive to the selected bandwidth (as discussed in Problems 6 and 7), leading to an asymmetric

distribution of Plan D contracts above and below the threshold (Problem 9). Additionally, the 3-star ratings undergo the most significant number of round-ups compared to higher-rated plans, making that area particularly volatile for analysis.